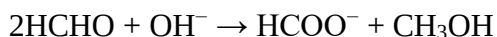


ASSIGNMENT (Equivalent Concept)

1. A volume of 1.12 l dry chlorine gas at 0°C and 1 atm was passed over a heated metal when 5.55 g of chloride of the metal was formed. If the atomic mass of the metal is 40, its valency is
2. HCHO disproportionates to HCOO^- and CH_3OH in the presence of OH^- (Cannizzaro's reaction).



If the equivalent weight of HCHO is E, then the value of E/10 is

3. When a solid element is reacted with chlorine, a gaseous chloride of vapour density 68.75 is formed. If this reaction is performed at constant temperature and pressure, the volume of the system reduces by one third. If the equivalent weight of the solid element is E, then the value of $\left(\frac{12}{31} \times E\right)$ is
4. V litre of SO_2 at 0°C and 1 atm is required to reduce 16.9 g of HClO_3 to HCl . The number of moles in '5 V' litre of SO_2 at 273°C and 2 atm is
5. The approximate mass (in g) of N_2H_4 can be oxidized by 24 g of K_2CrO_4 is (Cr = 52)

$$3\text{N}_2\text{H}_4 + 4\text{CrO}_4^{2-} + 4\text{H}_2\text{O} \rightarrow 3\text{N}_2 + \text{Cr}(\text{OH})_4^- + 4\text{OH}^-$$
6. A sample of pure $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ requires 30 mol of NaOH for titration. How many moles of KMnO_4 will the same sample react with, in acid medium?
7. A quantity of 1.245 g of $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$ was dissolved in water and H_2S was passed into it till CuS was completely precipitated. The H_2SO_4 produced in the filtrate required 10 ml of M – NaOH solution. Calculate x.
8. A mixture of CS_2 and H_2S when oxidized, yields a mixture of CO_2 , SO_2 and $\text{H}_2\text{O}(\text{g})$, which exerts a pressure of 7.2 atm, when collected in 82.1 L vessel at 327°C. To oxidize SO_2 in the mixture, 5 L of 2 N-iodine was required. Moles of CS_2 in the mixture is
9. One litre of a sample of ozonized oxygen at 0°C and 1 atm on passing through a KI solution, liberated iodine which required 9 ml of a thiosulphate solution. A volume of 12 ml of a '5.6 volume' hydrogen peroxide solution liberated iodine from another iodide

solution, which required 24 ml of the same thiosulphate solution. The volume per cent of ozone in the ozonized oxygen sample is

10. A certain mass of anhydrous oxalic acid is converted into H_2O , CO_2 and CO , on heating in the presence of H_2SO_4 . The CO formed reacts completely with iodine pentoxide to liberate iodine. The iodine thus liberated required 200 ml of 0.2 N thiosulphate. The mass (in g) of oxalic acid taken was
11. A quantity of 1.16 g $\text{CH}_3(\text{CH}_2)_n\text{COOH}$ was burnt in excess of air and the resultant gases (CO_2 and H_2O) were passed through excess NaOH solution. The resulting solution was divided into two equal parts. One part requires 50 ml of N – HCl for neutralization using phenolphthalein indicator. Another part required 80 ml of N – HCl for neutralization using methyl orange indicator. The value of n is
12. An amount of 0.1 moles of OH^- ions is obtained from 8.50 g of hydroxide of a metal. What is the equivalent weight of the metal?
13. A quantity of 2.7 g of an alloy of copper and silver was dissolved in moderately conc. HNO_3 and excess of HCl was added to this solution when 2.87 g of a dry precipitate is formed. Calculate the percentage of copper in the alloy. ($\text{Cu} = 63.5$, $\text{Ag} = 108$)
14. Peroxides, like oxides, are basic. They form hydrogen peroxide upon treatment with an acid. What volume (in ml) of 0.25 M – H_2SO_4 solution is required to neutralize a solution that contains 7.2 g of CaO_2 ?
15. A volume of 30 ml of a solution containing 9.15 g per litre of an oxalate $\text{K}_x\text{H}_y(\text{C}_2\text{O}_4)_z \cdot n\text{H}_2\text{O}$ is required for titrating 27 ml of 0.12 N – NaOH and 36 ml of 0.12 N – KMnO_4 separately. Assume all H-atoms are replaceable and x, y and z are in the simple ratio of g-atoms. The value of $xyzn$ is
16. The saponification number of fat or oil is defined as the number of mg of KOH required to saponify 1 g oil or fat. A sample of peanut oil weighing 1.5 g is added to 25.0 ml of 0.4 M – KOH . After saponification is complete, 8.0 ml of 0.25 M – H_2SO_4 is needed to neutralize excess of KOH . What is the saponification number of peanut oil?
17. A quantity of 1.6 g of pyrolusite ore was treated with 50 ml of 1.0 N-oxalic acid and some sulphuric acid. The oxalic acid left undecomposed was raised to 250 ml in a flask. A volume of 25 ml of this solution when titrated with 0.1 N – KMnO_4 required 32 ml of the solution. The percentage of available oxygen in the ore is

18. Calculate the amount (in mg) of SeO_3^{2-} in solution on the basis of the following data: 20 ml of M/40 solution of KBrO_3 was added to a definite volume of SeO_3^{2-} solution. The bromine evolved was removed by boiling and excess of KBrO_3 was back titrated with 7.5 ml of M/25 solution of NaAsO_2 . The reactions are (Se = 79)
- $$\text{SeO}_3^{2-} + \text{BrO}_3^- + \text{H}^+ \rightarrow \text{SeO}_4^{2-} + \text{Br}_2 + \text{H}_2\text{O}$$
- $$\text{BrO}_3^- + \text{AsO}_2^- + \text{H}_2\text{O} \rightarrow \text{Br}^- + \text{AsO}_4^{3-} + \text{H}^+$$
19. 7.5 ml of a hydrocarbon gas was exploded with excess of oxygen. On cooling it was found to have undergone a contraction of 15 ml. If the vapour density of the hydrocarbon is 14, determine its molecular formula. (C = 12, H = 1.)
20. In the analysis of 0.500 g sample of feldspar, a mixture of the chlorides of sodium and potassium is obtained, which weight 0.1180 g. subsequent treatment of the mixed chlorides with silver nitrate gives 0.2451 g of silver chloride. What is the percentage of sodium oxide and potassium oxide in feldspar?
21. 50 ml of mixture of CO and CH_4 was exploded with 85 ml of O_2 . The volume of CO_2 produced was 50 ml. Calculate the percentage composition of gaseous mixture, if all volumes are measured under same condition of temperature and pressure.
22. 500 ml of 2 m CH_3COOH solution is mixed with 600 ml 12% w/v CH_3COOH solution then calculate the final molarity of solution.
23. In order to remove Pb^{2+} from 10 litre H_2O , $\text{Na}_2\text{H}_2\text{EDTA}$ (0.4 M, 100 mL) is required.
- $$\text{PbCl}_2(\text{aq}) + \text{Na}_2\text{H}_2\text{EDTA} \rightarrow 2\text{NaCl} + \text{PbH}_2\text{EDTA}$$
- Hence millimoles of PbCl_2 present in 1 litre of H_2O is:
24. 150 ml of an aq. solution containing 5 millimoles of A (specific gravity = 1.2) is mixed with 250 ml of another aq. solution containing 10 millimoles of A (specific gravity = 1.4). If on mixing the density of the solution becomes $\frac{5.3}{4.5}$ gm / mL, then the molarity of A in the final solution becomes x M. The value of x is:
25. Calculate the molar ratio in which the following two substances would react?
- $\text{Ba}_3(\text{PO}_4)_2$ and AlCl_3